Bhabha scattering 2

Let $\mathcal{M}_1$ be the amplitude for $1 \to 3$, $4 \to 2$.

$$\mathcal{M}_1 = \iint \bar{\psi}_2(y)(-ie\gamma^\mu)\psi_4(y)G(x,y)\bar{\psi}_3(x)(-ie\gamma_\mu)\psi_1(x)\,d^4x\,d^4y$$

Function $G(x,y)$ is the photon propagator

$$G(x,y) = \frac{1}{(2\pi)^4}\int \frac{\exp(-ip\cdot(x-y))}{p^2+i\epsilon}\,d^4p$$

The wave functions are

$$\begin{aligned}
\psi_1(x) &= \sqrt{2EV}u_1\exp(-ip_1\cdot x)\\
\psi_2(x) &= \sqrt{2EV}v_2\exp(ip_2\cdot x)\\
\psi_3(x) &= \sqrt{2EV}u_3\exp(-ip_3\cdot x)\\
\psi_4(x) &= \sqrt{2EV}v_4\exp(ip_4\cdot x)
\end{aligned}$$

where V is a normalization volume.

Substitute for the wave functions in $\mathcal{M}_1$ and factor the spinor products.

$$\mathcal{M}_1 = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1) \iint \exp(i(-p_2+p_4)\cdot y)\,G(x,y)\exp(i(p_3-p_1)\cdot x)\,d^4x\,d^4y$$

Substitute for $G(x,y)$.

$$\mathcal{M}_1 = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1) \frac{1}{(2\pi)^4}\iiint \exp(i(-p_2+p_4)\cdot y)\,\frac{\exp(-ip\cdot(x-y))}{p^2+i\epsilon}\exp(i(p_3-p_1)\cdot x)\,d^4x\,d^4y\,d^4p$$

Combine exponentials.

$$\mathcal{M}_1 = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1) \frac{1}{(2\pi)^4}\iiint \frac{\exp(i(-p_2+p_4+p)\cdot y)\exp(i(p_3-p_1-p)\cdot x)}{p^2+i\epsilon}\,d^4x\,d^4y\,d^4p$$

Solve the integral over x.

$$\int \exp(i(p_3-p_1-p)\cdot x)\,d^4x = (2\pi)^4\delta^4(p_3-p_1-p)$$

Solve the integral over y.

$$\int \exp(i(-p_2 + p_4 + p) \cdot y)\, d^4y = (2\pi)^4 \delta^4(-p_2 + p_4 + p)$$

Hence

$$\mathcal{M}_1 = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1)(2\pi)^4 \int \frac{\delta^4(-p_2 + p_4 + p)\delta^4(p_3 - p_1 - p)}{p^2 + i\epsilon}\, d^4p$$

The integrand vanishes except for

$$p = p_2 - p_4 = p_3 - p_1$$

Hence

$$\mathcal{M}_1 = -\frac{e^2}{(2EV)^2}(\bar{v}_2\gamma^\mu v_4)(\bar{u}_3\gamma_\mu u_1)(2\pi)^4 \frac{\delta^4(p_4 + p_3 - p_2 - p_1)}{(p_3 - p_1)^2}$$

Let $\mathcal{M}_2$ be the amplitude for $1 \to 2$, $4 \to 3$.

$$\mathcal{M}_2 = \iint \bar{\psi}_3(y)(-ie\gamma^\mu)\psi_4(y)G(x,y)\bar{\psi}_2(x)(-ie\gamma_\mu)\psi_1(x)\, d^4x\, d^4y$$

It follows that

$$\mathcal{M}_2 = -\frac{e^2}{(2EV)^2}(\bar{u}_3\gamma^\mu v_4)(\bar{v}_2\gamma_\mu u_1)(2\pi)^4 \int \frac{\delta^4(p_3 + p_4 + p)\delta^4(-p_2 - p_1 - p)}{p^2 + i\epsilon}\, d^4p$$

The integrand vanishes except for

$$p = -p_3 - p_4 = -p_1 - p_2$$

Hence

$$\mathcal{M}_2 = -\frac{e^2}{(2EV)^2}(\bar{u}_3\gamma^\mu v_4)(\bar{v}_2\gamma_\mu u_1)(2\pi)^4 \frac{\delta^4(p_4 + p_3 - p_2 - p_1)}{(p_1 + p_2)^2}$$

Eigenmath script